DESIGN MILESTONES

Discover

Problem Exploration

- Research notes
- Insight diagram
- Scenarios
- Persona

Clarifier:
#problemDefinition
#designBrief
#processClarification

Define

Problem Definition

- Design Frame
- Concept Direction

Re-framer:
#experienceReframing
#designStrategy
#ritualPerspective

Develop

Solution Exploration

- Sketches
- Prototype
- Mechanism Diagram
- App Design
- Testing

Expander:
#conceptIdeation
#featureGeneration
#mechanismIdeas
#visualPrompt
Critique

Deliver

Final Solution

- Final Product

Visualization:
#visualCreation
#aiRenderingSupport
#storyboard

1. DISCOVER

Wanting to repeat same recipe

Social media content about caffeine effects

Dedicated preparation space

Ritual-oriented slow living trend

Caffee Corner

Siphon Brewer: The hottest coffee you can get.

Coffee a way to reward oneself

Coffee functions both as personal retreat and shared bonding moment.

Coffee and water quantity are important.

Temperature is important.

Can the equipment used be reduced?

Double bloom releases more CO_2.

Swirling helps the bloom increase

When the bloom increases, the taste of the coffee gets better.

Handling of coffee waste

USE SCENARIO: GUEST MODE

1. Intention
- Offer coffee
- Initiate connection

Emotion: Social Anticipation

2. Preparation
- Calculate servings
- Prepare cups
- Curate presentation
- Select bean type in app
- Set brewing parameters

Emotion: Care + Subtle Performance

3. Heating
- Machine heats water automatically
- Background ritual sound

Emotion: Shared Pause

4. Brewing
- Explain process
- Observe bloom phase
- Shared observation

Emotion: Pride + Engagement

5. Serving
- Hand cups
- Observe reactions
- Wait for feedback

Emotion: Social Relief / Shared Satisfaction

6. Aftermath
- Clear space
- Sustain warmth of interaction

Emotion: Social Closure

Age: 32
Profession: Remote worker
Skill: Medium brewing knowledge
Time Pressure: Moderate
Living: Alone, frequently hosts friends

Behaviors
- Repeat the same recipe
- Follows expert methods (bloom, timing)
- Notices sound, temperature, cooling
- Uses digital recipes/content
- Prepares coffee for guests

Motivations
- Daily rhythm structuring
- Sensory satisfaction
- Taste precision
- Emotional attachment to objects
- Care & bonding through hosting

Mindset
- Ritual-oriented
- Slow living influence
- Cultural awareness (fika)
- Sustainability-conscious

Tensions
- Consistency vs time pressure
- Ideal drinking temperature
- Waste awareness
- Ritual depth vs convenience

Users enjoy the ritual of brewing coffee but struggle to maintain consistency without slowing down their social interactions.

What I did:
- I researched coffee brewing rituals.
- I analyzed the manual pour-over experience.
- I identified the problems users encountered.
- I reviewed trend reports.

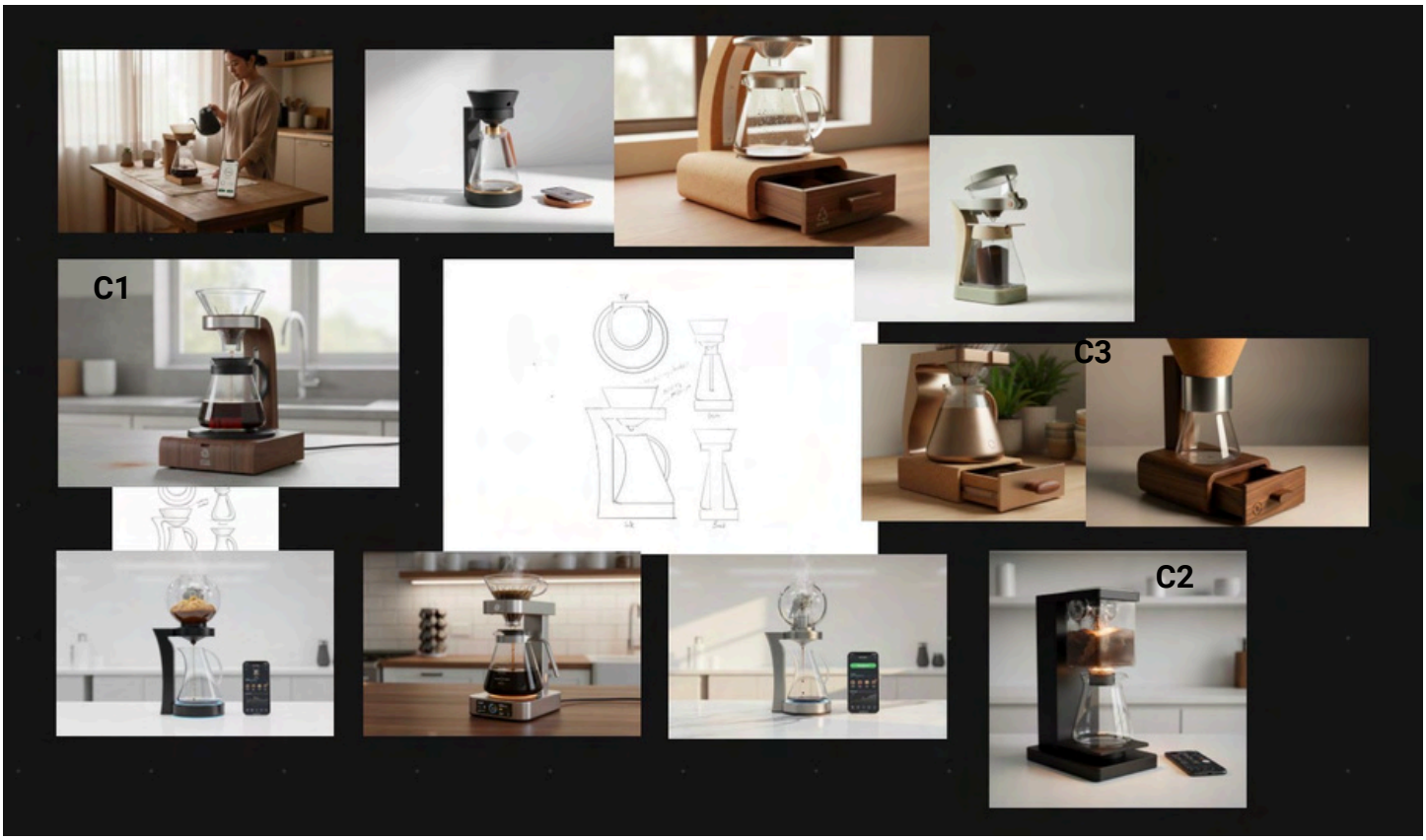
Insights I found:
- Users don't want to lose their coffee brewing ritual.
- But they want consistent quality and convenience.
- Bloom process and temperature control are often done incorrectly.
- Heat loss affects coffee quality.

Help I received from AI:
- Research on coffee brewing systems.
- List of futuristic coffee brewing devices.
- Extracting the common intersection of trend reports.
- I used it to create personas and journey maps.

2. DEFINE

Problem Definition
How might we help users brew high-quality coffee consistently without removing the ritual of manual brewing?

Design Brief
A smart coffee brewing system that supports the ritual instead of replacing it, while ensuring optimal brewing parameters.



CONCEPT 1
Direction: Manual ritual + subtle stabilization
Differentiation

- Rocking bloom support
- Heat retention base
- Manual pouring preserved
- No device interface
- App-only tracking
- Stability without automation
- Ritual integrity first

CONCEPT 2
Direction: Subtle stabilization + system automation
Differentiation

- Rocking bloom
- Heat stabilization
- Automatically draws water from the reservoir
- Bean selection via app
- Preset optimization
- Reduced cognitive load
- Precision support over pure ritual

CONCEPT 3
Direction: Manual ritual + sustainability loop
Differentiation

- Rocking bloom
- Heat retention
- Grounds collection compartment
- Waste tracking via app
- Sustainability without guilt
- Meaning beyond brewing
- Upcycling network

Help I received from AI:

- I visualized my concepts using vizcom AI, using a sketch I had made to see what they might look like.

Opportunity Areas:
Maintaining the same taste even under time pressure
Reducing cognitive load with guidance
Reducing noise by manually adding hot water
Automatic calibration for better flowering
Repeated ritual with a single touch
Circular feedback (app integration)

Tension 1
Need for a conscious pause (Acceleration in the city)
Desire to slow down
Reality of time pressure

Tension 2
Sensitivity = Cognitive Load
Wanting consistent results
Avoiding technical thinking

Brief rationale
It will enable even people with moderate knowledge of coffee to enjoy consistent, delicious coffee.
It will provide emotional satisfaction during the brewing process by making the process visible.
City life will not slow down, but it will be possible to capture the same feeling and experience emotional slowing down.
Psychological slowing down will become more valuable.
Ritual-focused design creates not only functionality but also connection.

3. DEVELOP

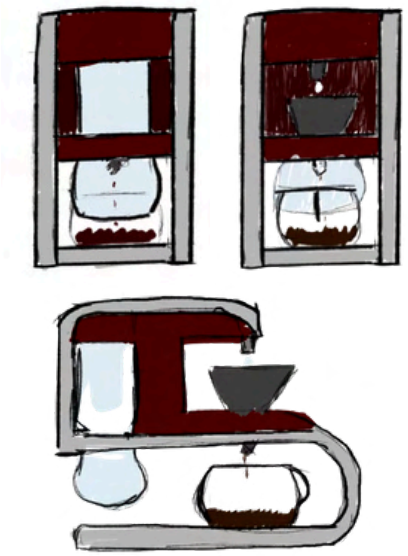
I will design a preparation system that preserves personal brewing rituals while quietly ensuring repeatable quality and transforming daily coffee into a meaningful, low-friction, sustainable act.

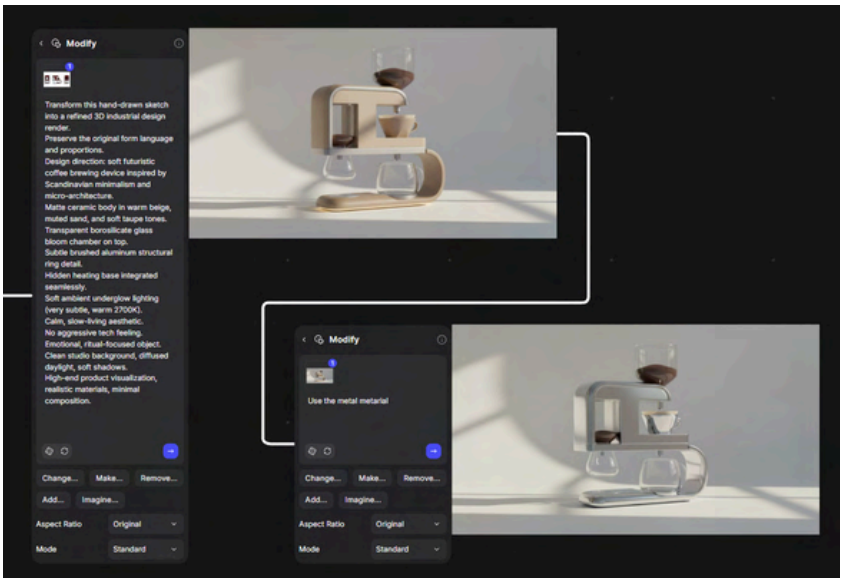
I decided to keep the coffee grounds separate from the machine because collecting them in the container underneath the device might raise hygiene concerns. I chose Concept 2, which differs from Concept 1 in that the machine draws water from the container and determines the ideal temperature according to the coffee beans. Since people expect the machine to also heat the water when they buy it, and to adjust the water temperature via the app to improve the flavor, having the water drawn from the machine itself will be better for the product.

Help I received from AI:

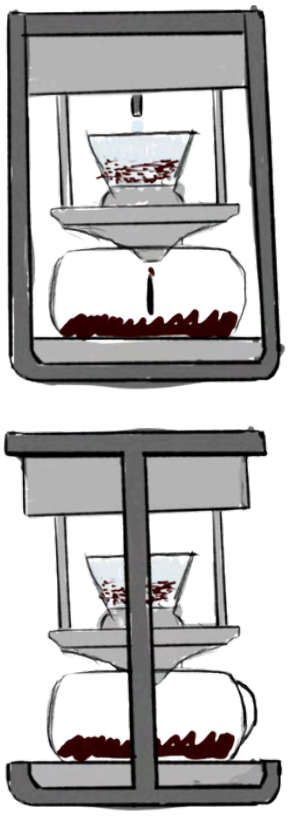
- I visualized my concept using vizcom AI, using sketches I had made to see what they might look like.
- Based on the specifications I decided on, I used AI to determine which technical components I needed for the device.

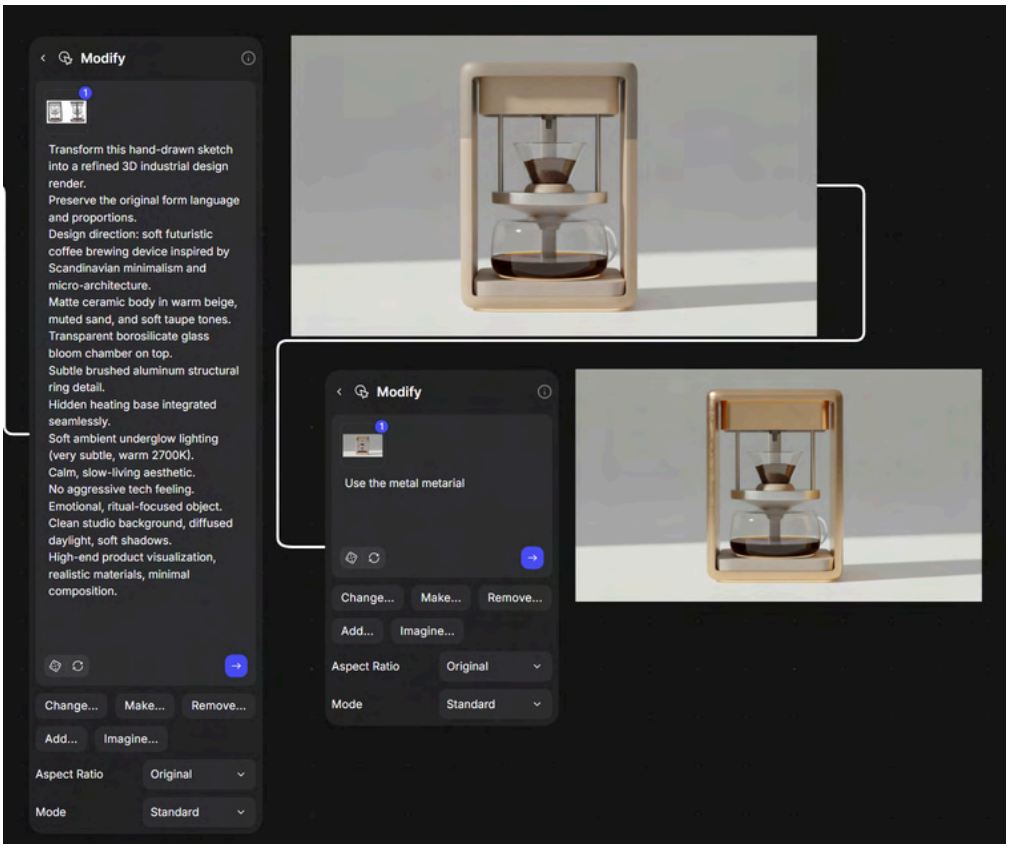
Option 1:



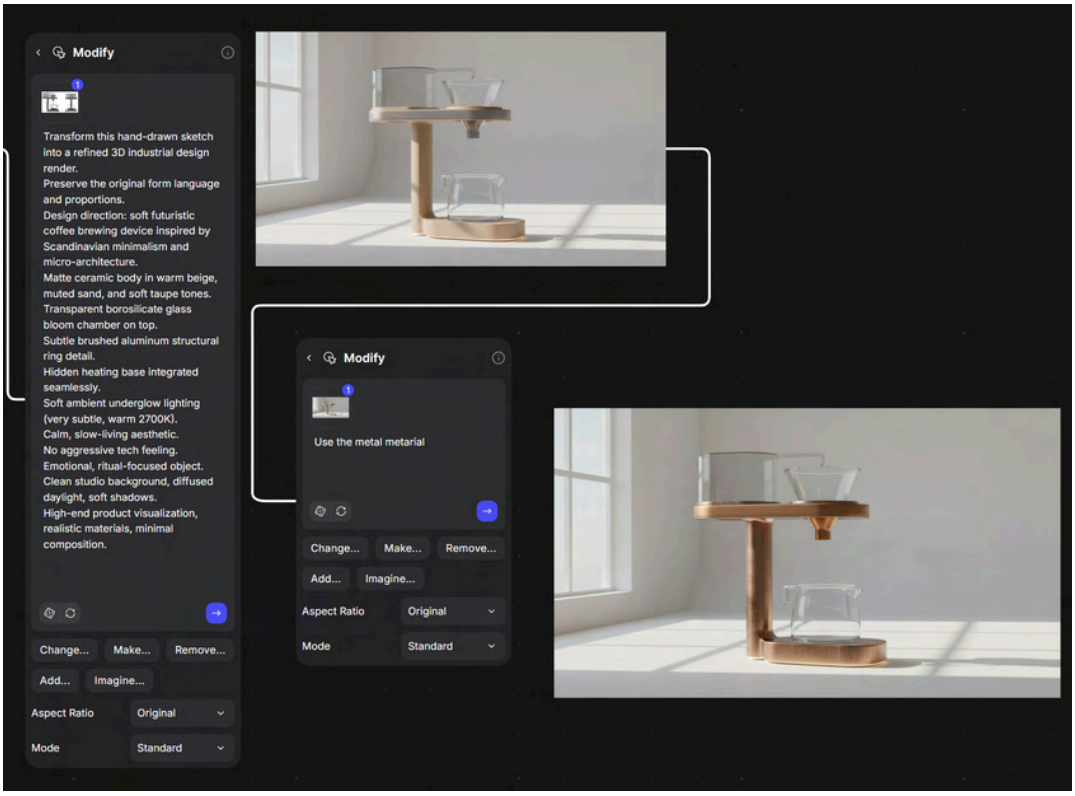


Option 2:



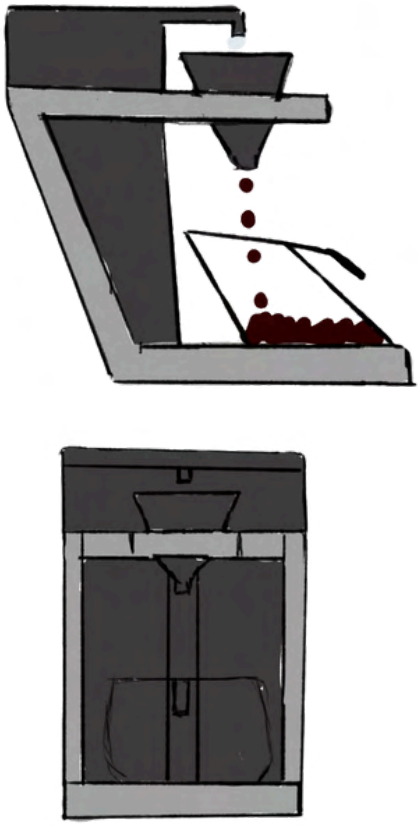


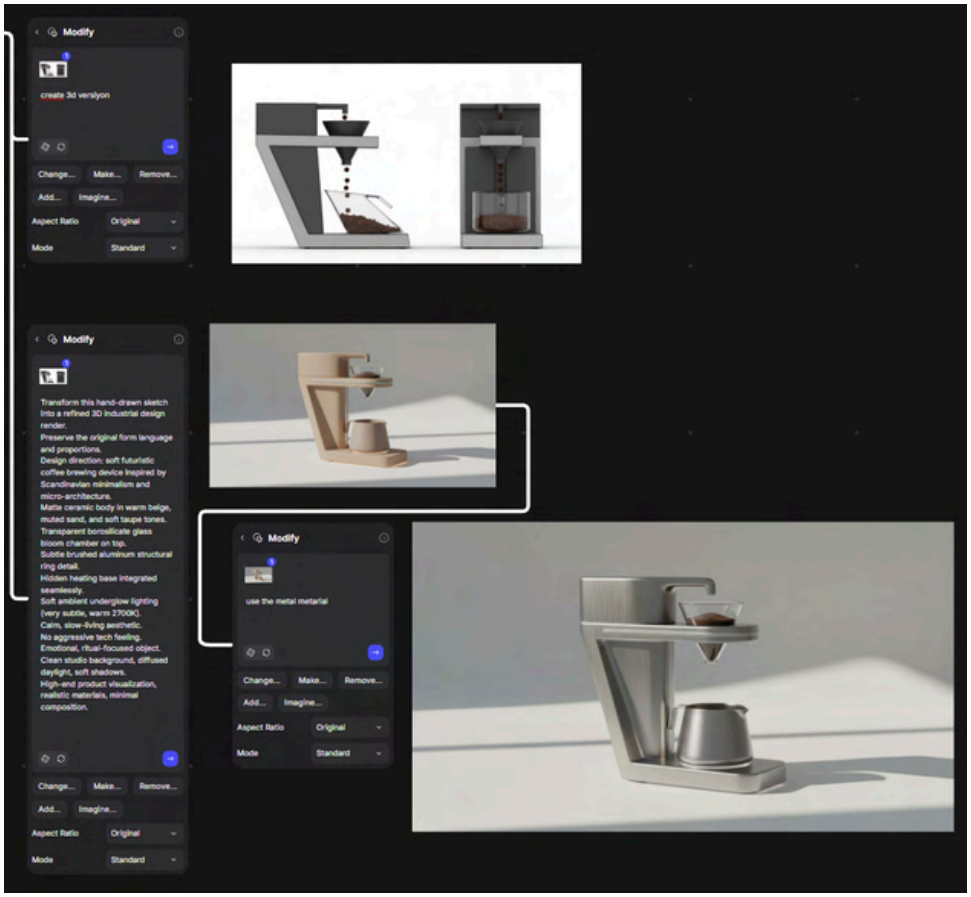
Option 3:



I decided to continue with Option 4.1, which fit the internal structure most easily and was more effective in terms of manufacturability, and I created its third version.

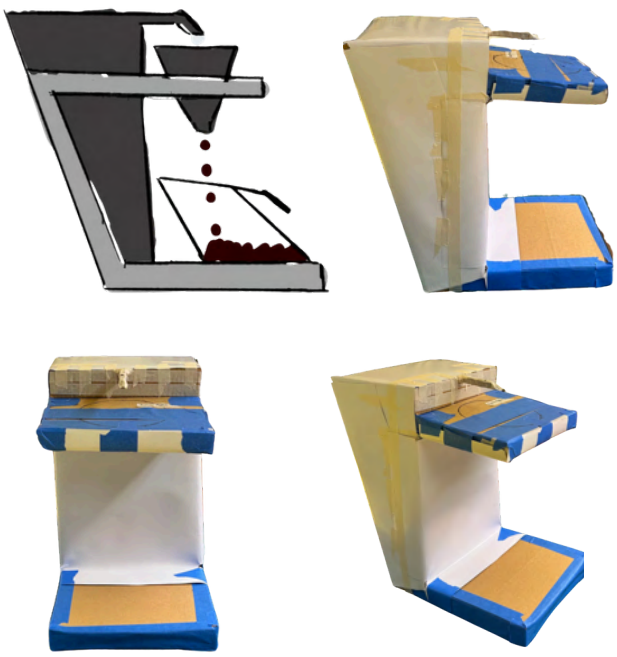
Option 4.v1:

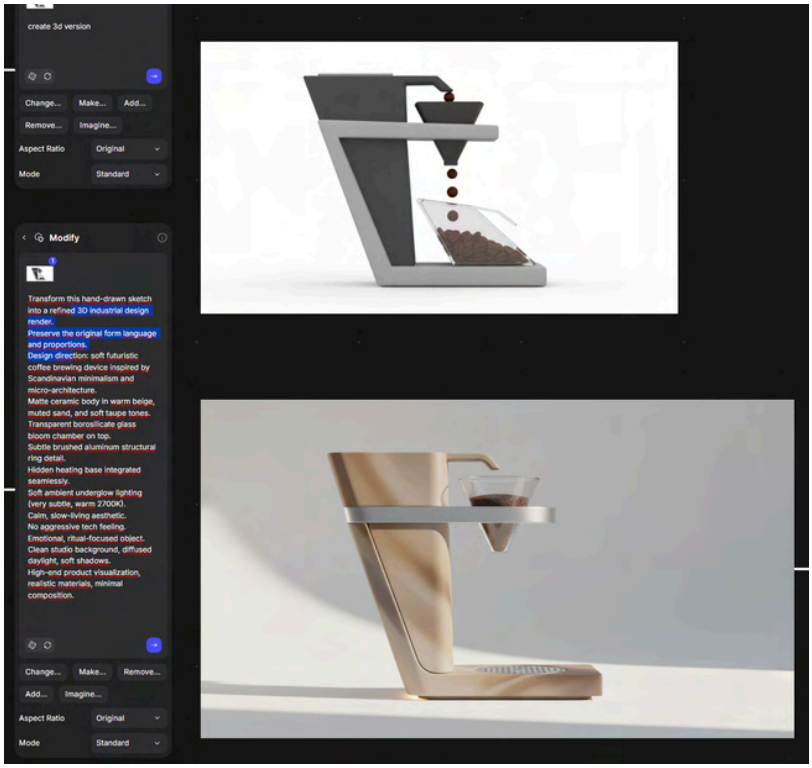




To continue the design language, I extended the top corner with a bevel and created sketch 4.2.

Option 4.v2:

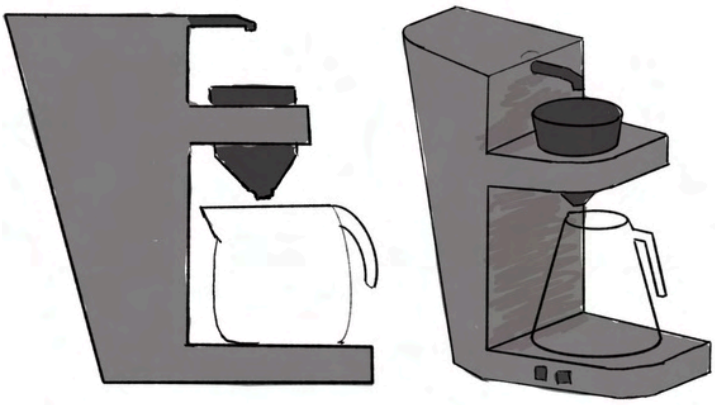


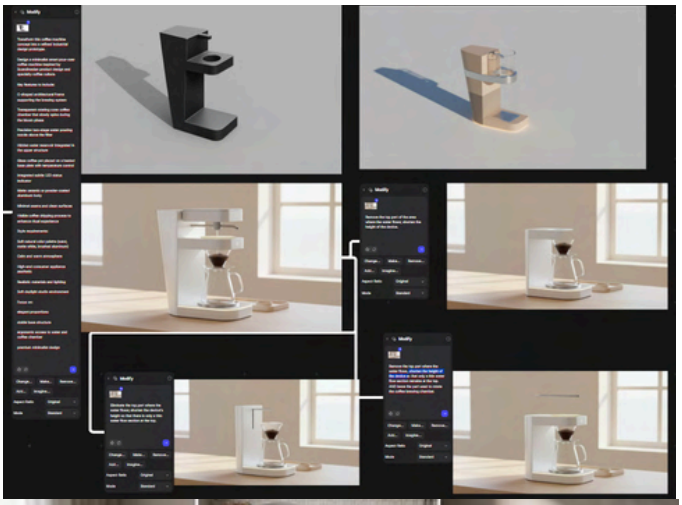


To accurately choose the product dimensions, I created a physical prototype. Based on the instructor's critique of the prototype, I made changes to the design. Since I wanted the product to be positioned sideways, I decided to break the symmetry on the front surface to create this perception. I tested it with AI using the sketches and prototype visuals. Based on the AI's suggestion, I decided to add a quick-brew button in addition to the on/off button, so the machine can still be used if the app has issues. I believe that placing these buttons on the front will help the user understand the orientation of the device. Based on all of this, I created the 4.3 sketch.

4. DELIVER

Option 4.v3:



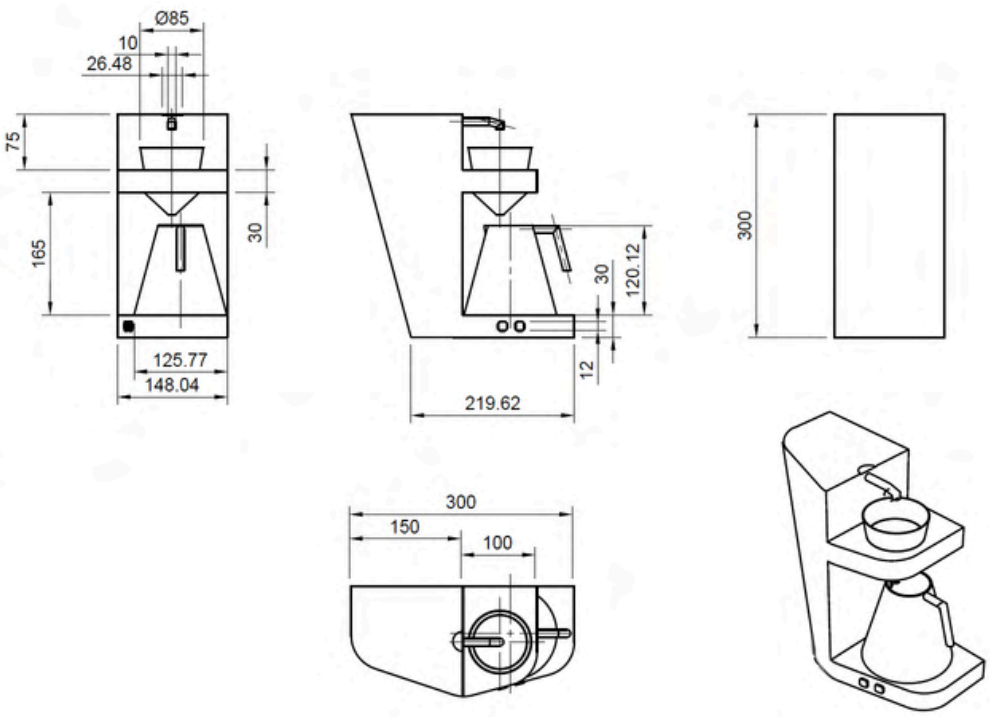


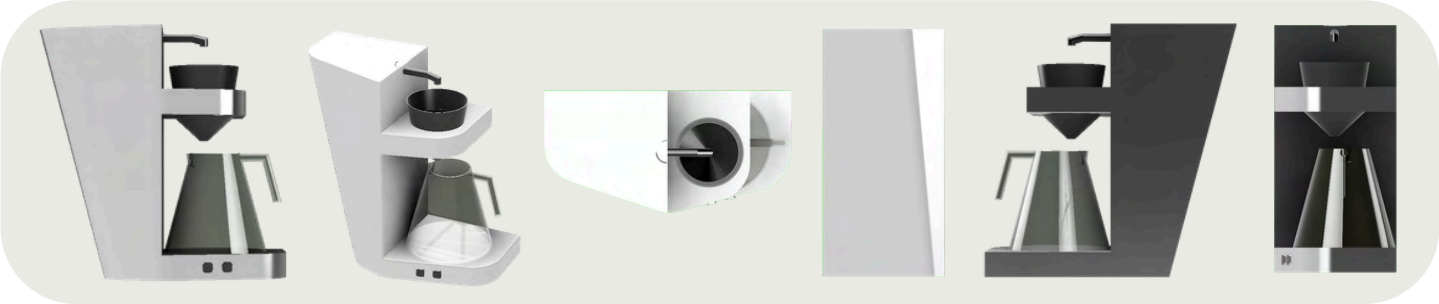
Concept Statement

This project proposes a coffee brewing system that preserves the ritual of manual brewing while subtly supporting consistency and quality through intelligent assistance. The device heats the water to the optimal temperature for the selected coffee beans and dispenses it in two stages to ensure proper brewing and extraction. The slow horizontal rotation of the coffee chamber gently increases the brewing, reducing the need for manual stirring, while preserving the visual experience of brewing and thus enhancing the flavor of the coffee by supporting bloom. The system works in conjunction with a mobile application where users can select bean types, save previous brewing parameters, and monitor caffeine consumption. By balancing automation and ritual, the design reduces friction in the brewing process while respecting the emotional value of manual coffee preparation. In addition, the system offers a sustainable layer by allowing users to donate their collected coffee grounds through an integrated network.

Materials List:

- Body: Powder-coated stainless steel for a premium and durable outer shell.
- Dripper: Food-grade Tritan to provide heat resistance, light weight, and support the rotating mechanism.
- Pot (Carafe): Borosilicate glass for thermal resistance and visual clarity during brewing.





System Diagram (User – App – Device)

USER

Inputs:

- Ground coffee is placed in the brewing chamber
- Water is filled in the reservoir
- User selects coffee bean type in the app
- User presses Start Brewing
- User decides whether to donate coffee grounds

MOBILE APP

Processes

- Receives bean selection from the user
- Sends brewing parameters to the device
  - ideal water temperature
  - water amount
  - brewing timing
- Records brewing data
- Calculates caffeine intake
- Shows brewing progress
- Displays coffee grounds donation option

DEVICE

Processes

- Receives bean selection from the user
- Sends brewing parameters to the device
  - ideal water temperature
  - water amount
  - brewing timing
- Records brewing data
- Calculates caffeine intake
- Shows brewing progress
- Displays coffee grounds donation option

OUTPUTS

- Fresh brewed coffee
- Warm coffee pot
- Brewing feedback in app
- Brewing history
- Caffeine consumption tracking
- Coffee grounds available for donation network

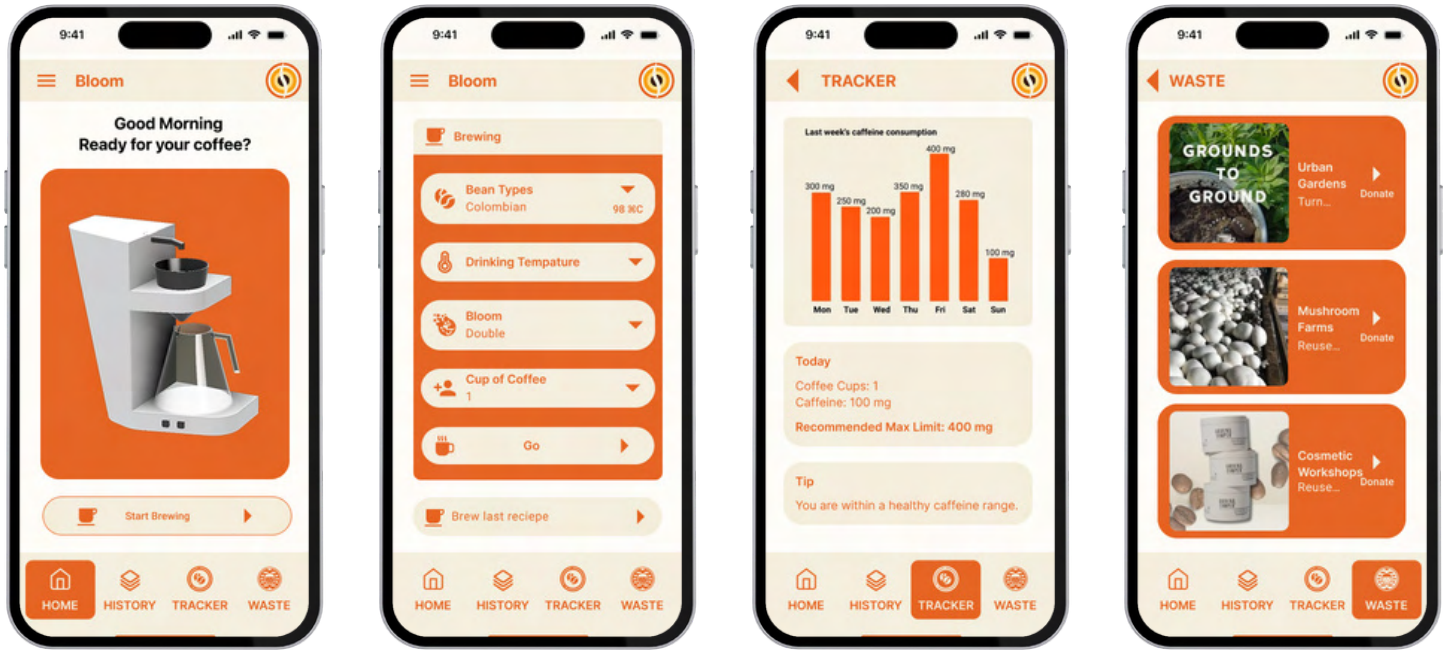
SUSTAINABILITY LOOP

- Coffee Grounds
- App notifies collection network

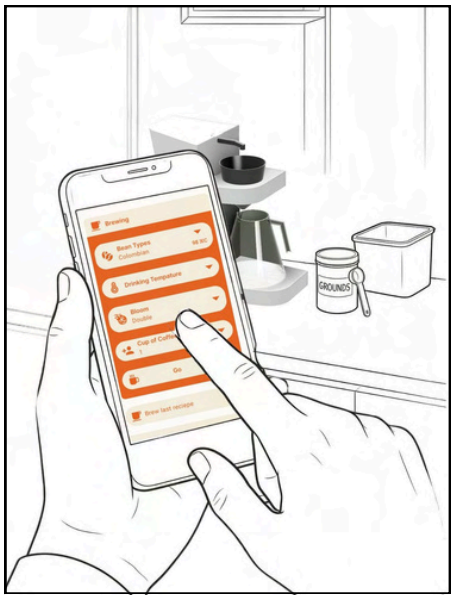
Grounds reused for:

- Compost, mushroom growing, bio material

Interface Design and User Scenario



1- User adds ground coffee to the brewing chamber.



2- User selects settings in the app and starts brewing.

3- The device heats water and pours it over the coffee.

4- The chamber rotates gently to support extraction.

5- Fresh coffee flows into the pot.

6- The user enjoys a fresh cup of coffee.